Litao LIU

Phone: +86 153 4053 3199
Email: liulitao011026@gmail.com
Homepage: https://litaoliu01.github.io/
Birthdate: October 26, 2000, in Chongqing, China

EDUCATION

Institute of Automation, Chinese Academy of Sciences (CASIA) **Beijing, China**
Research Intern in Robot Learning, Advisor: Wenzhao Lian *11/2023-10/2024*

Sichuan University - Pittsburgh Institute (SCUPI) **Chengdu, China**
Bachelor of Engineering in Mechanical Engineering *09/2019-06/2023*

- **GPA:** 3.85/4.00, 91.54/100 (**Top 4** of 65)

Sichuan University - The Hong Kong Polytechnic University Institute for Disaster Management and Reconstruction (IDMR) **Chengdu, China** *09/2020-06/2023*
International Disaster Reduction and Emergency Management Innovation Class

PUBLICATIONS

- **Litao Liu**, Wentao Wang, Yifan Han, Zhuoli Xie, Pengfei Yi, Juyan Li, Yi Qin, Wenzhao Lian. "FoAM: Foresight-Augmented Multi-Task Imitation Policy for Robotic Manipulation." *arXiv preprint arXiv:2409.19528 (2024)*. https://projfoam.github.io/ (Under review for the 2025 IEEE International Conference on Robotics and Automation, ICRA 2025; decision expected by January 2025)
- Pengfei Yi, Yifan Han, Junyan Li, **Litao Liu**, Wenzhao Lian. "Viewpoint Matters: Dynamically Optimizing Viewpoints with Masked Autoencoder for Visual Manipulation." (Under review for the 2025 IEEE International Conference on Robotics and Automation, ICRA 2025)
- **Litao Liu**, Bingwei Tian. "Comprehensive Engineering Frequency Domain Analysis and Vibration Suppression of Flexible Aircraft Based on Active Disturbance Rejection Controller." *Sensors*, 22.16 (2022): 6207. https://doi.org/10.3390/s22166207 (SCI, IF=3.85, JCR Q2)
- **Litao Liu.** "Stock Investment and Trading Strategy Model Based on Autoregressive Integrated Moving Average." *2022 IEEE Conference on Telecommunications, Optics and Computer Science (TOCS)*. IEEE, 2022. https://doi.org/10.1109/TOCS56154.2022.10015924
- **Litao Liu**, Lei Li, Haozhou Mo. "Research on Ecological Evaluation Model Based on Principal Component Analysis." *Proceedings of the 3rd International Conference on Green Energy, Environment and Sustainable Development (GEESD2022)*. Vol. 23. IOS Press, 2022. https://doi.org/10.3233/ATDE220312
- **Litao Liu**, Haozhou Mo. "Research on Raw Material Ordering and Transfer Strategy based on AHP-TOPSIS and 0-1 Planning." *Conference Proceedings of the 10th International Symposium on Project Management, China, ISPM 2022*, 1076-1083. https://doi.org/10.52202/065147-0146
- Yonggang Liang, Yuewei Ling, Yujie Li, **Litao Liu**, Nuo Xu. "Post-disaster air purification system based on bionic lung design." *International Conference on Automation Control, Algorithm, and Intelligent Bionics (ACAIB 2022)*, Vol. 12253. SPIE, 2022. https://doi.org/10.1117/12.2639442

RESEARCH EXPERIENCE

General Learning Policy with Intend Mechanism **Shanghai, China**
Guided by Prof. Wenzhao Lian | School of Artificial Intelligence, Shanghai Jiao Tong University *10/2024-Present*

- Develop a novel robot imitation learning framework that enables agents to learn effectively from large-scale, non-proprioceptive demonstrations. This framework enhances the adaptability of agents by leveraging diverse datasets to acquire versatile skills applicable across various tasks.
- Implement fine-grained sampling of agents' visual data to identify task operation intentions. These extracted intentions serve as generalizable skills, enabling agents to learn from diverse data domains and perform effectively in a variety of scenarios.

Exploration of Multi-Task Imitation Learning Policy **Beijing, China**
Guided by Prof. Wenzhao Lian | CASIA *03/2023-09/2024*

- Created the Foresight-Augmented Manipulation Policy (FoAM), an innovative multi-task imitation learning (MTIL) framework.
- Integrated a vision-language model as the Goal Imagination Module within FoAM, enabling the agent to leverage language prompts and initial visual observation to autonomously acquire multi-goal conditions, solve the limitations of single-condition policies, and generalize effectively to unseen tasks and scenarios.
- Proposed the Foresight-Augmented module, which learns the differences between current visual observations and corresponding goal images, significantly enhancing task activation success rates in multi-task learning.
- Developed, open-sourced, and actively maintained the FoAM benchmark featuring 10 task suites and over 80 tasks, accompanied by a dataset featuring 14 real-world tasks.
- Evaluated FoAM across 100+ tasks in both simulated and real-world environments, achieving up to a 41% improvement in success rate compared to state-of-the-art IL baselines.

Develop High-Performance Heat Exchanger **Chengdu, China**
Guided by Prof. Minking Chyu & Dr. Dong Liang | SCUPI *09/2022-09/2023*

- Designed and developed a bionically inspired heat exchanger (HX), the F-RD HX, tailored for enhanced heat transfer performance.
- Utilized ANSYS-ICEM and CFX to perform simulation experiments, uncovering the internal flow mechanisms and heat transfer characteristics of the F-RD HX.
- Conducted extensive real-world experimental studies to compare the F-RD HX with the state-of-the-art S-D HX. The results demonstrated that the F-RD achieves comparable heat transfer efficiency while significantly outperforming the S-D in mechanical properties, highlighting its potential as an ideal solution for applications requiring both superior heat transfer performance and mechanical durability.
- Explored hybrid bionic structures for HXs to improve design flexibility and enhance performance control.

Research on Vibration Suppression of Aircraft **Chengdu, China**
Guided by Prof. Bingwei Tian | IDMR *11/2021-08/2022*

- Designed a second-order active disturbance rejection controller to suppress the dynamics of an aircraft's fourth-order pitch angle vibration model.
- Developed a MATLAB-Simulink simulation program to conduct detailed simulation experiments and evaluate the controller's performance.
- Analyzed the closed-loop system's robustness, tracking response, and anti-disturbance capabilities. The Monte Carlo Shooting method was utilized to validate the controller's robust control performance under 3σ conditions, achieving a 99.7% confidence level. The results highlight the controller's strong potential for practical applications in aircraft vibration suppression.

Post-Disaster Air Management **Chengdu, China**
Guided by Prof. Tetsuo Shoji | IDMR & Tohoku University, Sendai, Japan *09/2020-10/2022*

- Designed a water-based air purifier inspired by the purification mechanism of mammalian lungs. This membrane-free, portable system is specifically suited for post-disaster rescue scenarios, providing clean air to victims. By mimicking the two-stage purification process in mammals, it effectively removes particulate matter such as PM2.5 and PM10, achieving a purification efficiency of up to 99.7%.
- Conducted comprehensive tasks, including market and policy research, technical documentation, patent application, experimental data measurement, and product performance evaluation.
- Established partnerships with urban emergency equipment companies to facilitate the adoption and deployment of the purifier in emergency scenarios.

Service Learning, Community Toughness and Resilience **Chengdu, China**
Guided by Prof. Bingwei Tian | IDMR *08/2021-3/2022*

- Conducted a comprehensive review of over 100 articles on community resilience and service learning, analyzing and summarizing key issues in the field.
- Analyzed and synthesized key insights, including the characteristics of service learning and actionable implementation goals for enhancing community resilience.
- Contributed to the editing and publication of a book, ensuring accuracy and clarity in presenting research findings.

INTERNSHIP & COMPETITIONS

Beijing CoreNetic Technology Co., Ltd., *CoreNetic.ai* **Beijing, China**

Dual-arm Robot Manipulation | Development Intern *11/2023-10/2024*

- Operated and applied a dual-arm robot in the field of drug grasping, aiming to enhance the intelligence of pharmacy operations.
- Utilized the Dynamic Movement Primitives (DMP) algorithm, which models and generates complex motion trajectories. Initially, smooth trajectory data is collected, and these complex trajectories are reproduced using the DMP algorithm. This approach enables the robot to perform consistent actions from different starting points, allowing it to combine various meta-actions to accomplish diverse tasks.

Shanghai Zhangxiaomen Education Technology Co., Ltd. **Shanghai, China**

High Scholl Theoretical Physics Teaching | Teaching Intern *08/2020-07/2021*

- Offered personalized instruction to high school students, focusing on building a solid understanding of complex physics concepts and supporting their academic growth. Conducted over ten hours of dynamic and interactive teaching sessions each week, tailored to address diverse learning needs. Furthermore, utilized the earnings from this internship to independently finance my undergraduate education, demonstrating both dedication and self-sufficiency.

Robot Scheme Design and Production Competition *04/2022*

- Designed and built an intelligent unmanned vehicle, utilizing an STM32 microcontroller as the central control unit. The vehicle was equipped with various modules, including grayscale sensors and visual equipment, enabling navigation and autonomous grasping capabilities.

Mathematical Contest in Modeling *02/2022*

- Developed and evaluated time series prediction models for Bitcoin and gold prices, utilizing Autoregressive Integrated Moving Average (ARIMA) and XGBoost neural networks. After comparing predictions against ground truth data, the ARIMA model was ultimately chosen for its superior performance in supporting effective trading decisions.

TEACHING ASSISTANT

- Class MEMS 0031: Electric Circuits, SCUPI *2023 Spring*
- Class ME 0071: Introduction to Fluid Mechanics, SCUPI *2022 Fall & 2023 Spring*
- Class ENGR0135: Statics and Mechanics of Material I, SCUPI *2022 Fall*
- Class ENGR0145: Statics and Mechanics of Material II, SCUPI *2022 Spring*
- Class MEMS0024: Introduction to Mechanical Design, SCUPI *2021 Fall & 2022 Fall*
- Class PHYS 0174: Physics for Science and Engineering I, SCUPI *2021 Spring*

ACTIVITIES & SERVICE

- Deputy Monitor & Study Committee, Class Council, IDMR *09/2020-06/2023*
- Deputy Monitor & Study Committee, Class Council, SCUPI *09/2021-06/2022*
- Vice President, International English Exchange Association, SCU *09/2020-09/2021*
- Volunteer, Shennongjia Forest District Culture and Tourism Bureau *07/2021*
- Delegate, Model United Nations Conference, SCU *04/2021-05/2021*
- Representative, The 32nd Student Congress, SCU *09/2020-09/2021*
- Student Ambassador, SCUPI *05/2020-06/2020*
- Volunteer, Return to High-school Social Practice of Winter Vacation, SCUPI *02/2020*
- Competition Coordinator, SCU Freshman Cup English Writing Competition 2019 *12/2019*
- International Spiritual Rescue Training, SCU *10/2019*

HONORS & AWARDS

- Outstanding Graduate, Sichuan Province, China **(Top 4%)** *2023*
- National Scholarship, China **(Top 1%)** *2022*
- National-level Project in 2022 Undergraduate Innovation and Entrepreneurship Training Program *2022*
- Outstanding Graduate, Sichuan University *2022*
- Outstanding Student Leader, Sichuan University *2022*
- Dean's List, SCUPI **(Top 10%)** *2022*

- Provincial Comprehensive Quality Assessment, Grade A, Sichuan Province *2022*
- First prize, The Second Emergency Science Popularization Competition of Disaster Prevention and Reduction, Sichuan Province *2022*
- Honorable Mention, 2022 Mathematical Contest in Modeling *2022*
- Outstanding Volunteer, Press of Chinese Journal of Health Psychology *2021*
- First Prize, "Chinese Tale – English Talk" Activity, SCUPI *2021*
- Excellent Student Scholarship, IDMR *2022 & 2021*
- Second Prize, 2020 China Disaster Prevention and Reduction Day - Emergency Escape and Evacuation Drawing Competition, IDMR *2020*

OTHER INFORMATION

- **Standardized Tests:** TOEFL 107 (R 30, L 29, S 21, W 27); GRE 337 (V 168, M 169) + AW 3.5
- **Programming Skills:** Proficient in Python, MATLAB, C, LaTeX; Advanced in C++, Java Script
- **Technical skills: Software:** MuJoCo, Unreal Engine, CATIA, SolidWorks, ANSYS, Adobe (Photoshop), Microsoft Office (Certificate of Level II), SPSS, Origin; **Hardware:** Xarm Robots, Universal-Robots, Micro Controller Unit (51 MCU, STM32), RealSense, Orbbec Camera
- **Hobbies:** Traveling, Listening to music while strolling, Reading, Badminton, K-pop, Cooking, Dancing

REFERENCES

- Prof. Wenzhao Lian, Professor, School of Artificial Intelligence, Shanghai Jiao Tong University, Shanghai, China, lianwenzhao@sjtu.edu.cn, Phone: +86 156 1105 9156
- Prof. Bingwei Tian, Associate Professor, Institute for Disaster Management and Reconstruction, Sichuan University - The Hong Kong Polytechnic University, Chengdu, China, bwtian@scu.edu.cn, Phone: +86 135 5120 0510
- Prof. Jin Xu, Associate Professor, Sichuan University - Pittsburgh Institute, Chengdu, China, jin.xu@scupi.cn, Phone: +86 198 2294 3021
- Dr. Dong Liang, Postdoctoral Researcher, Sichuan University - Pittsburgh Institute, Chengdu, China, dongliang@scupi.cn, medongl@ust.hk, Phone: +86 177 6133 3969, +852 9179 8630

Litao Liu

(English)

刘力滔

(Chinese)

November 29, 2024